SAMS Kernel

The F18A's display modes: every mode and option, with pictures

Version 1.34.0 · 28 October 2026
For a TI-99/4A with an F18A video card and a SAMS memory card
MIT License, © 2026 Rebecca Sutter

Contents

1. The modes at a glance

2. VMODE

3. The VRAM map

4. The glyph modes: Text 80, Text 40, Graphics I

5. Enhanced colour for tiles (VOTE1, VOTE2, VOTE3)

6. Graphics II (bitmap) and Multicolor

7. Sprites

8. Tile layer 2, hardware scrolling, page sizes

9. Palette

10. The bitmap layer

11. Other registers (set them with VREG)

12. What `VMODE` does not cover, and what I did not check

13. Trying the modes

The kernel boots in Text 80, but the F18A offers much more, and `VMODE` sets up any of it. This document is the reference for every mode and option the kernel sets, and for the registers beyond them (`VREG`): what each looks like, where its tables live, what the text routines do in it, and what to watch for. The screenshots are drawn by `tools/f18render.py` from the simulator running `examples/modes`, one screen per mode: that program is the worked example, and `tests/test_kernel.py modes` checks the pixels of each.

How far to trust this. The mode rules (which register selects what, how an attribute picks a palette entry, where sprites and layers draw) are those of js99er's F18A emulation, itself written from the F18A's documentation. I have not checked them against a real F18A, and parts the demo does not exercise (listed in part 12) I have not checked at all.

Routes. Two ways to work: [PC] / [TI] / [PC or TI]. [PC] is a PC with Python 3 and xdt99 (build with `skbuild.py`, run in the simulator or js99er); **[TI]** is the TI itself with the workbench cartridge (editor, assembler, launcher); **[PC or TI]** is the same either way. The full table is in PROGRAMMERS-GUIDE.md part 0. Here: `VMODE` and `VREG` calls are **[PC or TI]** code. **[TI]** `programs/mode.a99` on the workbench disk switches Text 80 / Text 40 (tested); any other mode is a different `VMODE` line in the same program, untested on the TI route. **[PC]** `examples/modes` shows all fifteen screens, and the pictures in this document are rendered on the PC (`tools/f18render.py`, `tools/mkshots.py`).

1. The modes at a glance

Five *base modes*, chosen by `VMODE`'s first argument, each in 24 or 30 rows (except Graphics II), with options for the colour of tiles and sprites. Every mode can show sprites; all but Graphics II and Multicolor can use enhanced colour for tiles.

VMODE	Mode	Screen	Tile	Colour comes from
`VMT80`	Text 80	80 x 24 or 30 characters	6 x 8	VR7 (two colours for the screen), or **one attribute byte per position** (`VOPOS`), or enhanced colour
`VMT40`	Text 40	40 x 24 or 30	6 x 8	as Text 80
`VMGM1`	Graphics I	32 x 24 or 30 tiles of 8 x 8 (256 x 192 or 240)	8 x 8	a colour table, one byte per 8 patterns (two colours); with enhanced colour, an attribute per tile or per position
`VMGM2`	Graphics II (bitmap)	256 x 192	8 x 8, three sets	a colour byte per 8 pixels of a pattern row (two colours)
`VMMC`	Multicolor	64 x 48 blocks of 4 x 4 pixels	2 x 2 blocks	the pattern table holds the block colours

Option (add to R1)	Meaning
`VO30`	30 rows (240 pixels high) instead of 24; not for Graphics II
`VOPOS`	an attribute byte for every screen position: text colours in Text 80/40, or the enhanced-colour attribute of each cell
`VOTE1`, `VOTE2`, `VOTE3`	enhanced colour for tiles: 1 plane (2 colours of a palette pair), 2 planes (4 colours), 3 planes (8 colours); Text 80/40 and Graphics I
`VOSE1`, `VOSE2`, `VOSE3`	enhanced colour for sprites, the same three levels
`VOSPR`	sprites on (VR51 = 32); without it the sprite list is stopped
`VOS16`	sprites 16 x 16 instead of 8 x 8
`VOSMG`	sprites magnified x2

Beyond these, the F18A has a **second tile layer** with its own tables and scrolling, a **bitmap layer** of 4 colours or 16 (fat pixels), hardware scrolling and page sizes, up to 32 sprites with linking, a 64-entry palette, a GPU and more. The kernel sets them to "off" in `VMODE` and gives you `VREG` to turn them on (parts 7, 8 and 10). `examples/modes` shows each.

2. VMODE

```
       LI   R0,VMT40             the base mode
       LI   R1,VO30+VOPOS        the options, added together
       BL   @VMODE
```

`VMODE` unlocks the F18A if it is not (the loader did), blanks the display, then:

1. writes VR0-VR7 for the mode (table in `video.a99`);
2. writes every register the options and the "extras" use: VR49 (rows, tile and sprite enhanced colour), VR50 (position attributes), VR51 (sprites), and, to their off values, VR10/11 (tile layer 2), VR19, VR24-VR31 (palette selects, scroll, page and plane sizes, sprite limit, bitmap layer), VR47-48;
3. sets the text routines' variables (below);
4. blanks the tables in the standard map (part 3): the name table, the attribute table where there is one (text colour on the page, or 0 for enhanced colour), the sprite attribute table, Graphics II's pattern and colour tables, and loads the **glyph set** into the pattern table of the three glyph modes (Text 80, Text 40, Graphics I); for enhanced colour 2 and 3 it also clears the extra pattern planes;
5. installs the DMA mover's GPU program and loads the current colour scheme into palette entries 0-15 (`LDSCHM`);
6. turns the display on and sets `VR1CPY`.

A mode and option pair a mode cannot use are dropped: Graphics II and Multicolor ignore `VOPOS` and `VOTE1-3`; Graphics II ignores `VO30`. It does not touch sprite *patterns* or sprite *colours* (those are yours), slot B (it puts it back), or the file service.

Variables for the text routines (set by `VMODE`; read them, do not write them):

Variable	Holds
`VMNUM`, `VMOPT`	the mode and options last set
`TXCOLS`, `TXROWS`, `TXCELLS`	columns, rows, and their product: 80/40/32, 24/30
`TXATTR`	non-zero when the mode has an attribute table the text routines write (Text 80/40 with `VOPOS` or enhanced colour; Graphics I with enhanced colour)
`NAMBAS`	the name table's VRAM address (`NAMTAB`, or `NAMGM2` in Graphics II)

The boot mode, `T80INI`, is `VMODE` with `VMT80` and `VMDEF` (= `VO30+VOPOS`), then the DMA test. `COLS` and `ROWS` stay as constants for the boot mode (80 and 30); code that must follow the mode uses `TXCOLS` and `TXROWS`.

`VREG` writes any F18A register: `R0` = its number (0-63), `R1` = the value in the high byte: `LI R0,VR_HS2 / LI R1,>0300 / BL @VREG`. The register names (`VR_NAME2`, `VR_BMPCT`, ...) are in `kdefs.a99`.

A mode switch rewrites the screen. `VMODE` is a full re-initialisation: everything on the screen is gone afterwards, and so are tables you loaded into the standard places. Redraw after it, from your own state. It takes a moment (Graphics II clears 12K of VRAM). Switching between Text 80 and Text 40 loses text; the variables and your slot A data are untouched. `F18OFF` (the quit path) puts back every register `VMODE` and `VREG` can have changed.

3. The VRAM map

All modes but Graphics II share one map, arranged so that the enhanced colour modes (pattern planes `>800` apart for tiles, `>200` apart for sprites) fit beside the sprites and the file service:

Range	Holds	Equate
`>0000-095F`	name table (80 x 30 = 2,400 bytes; smaller in the other modes)	`NAMTAB`
`>0960-0967`	the GPU's mailbox	`GPUMB`
`>0970-098F`	the GPU's program	`GPUAD`
`>09C0-09DF`	Graphics I's colour table (32 bytes)	`GM1CT`
`>0A00-135F`	the attribute table (2,400 bytes at 80 x 30)	`ATRTAB`
`>1400-14FF`	the sprite attribute table (32 x 4; link bytes at +`>80`)	`SATAB`
`>1500-17FF`	**free** (768)	
`>1800-1FFF`	pattern table, plane 0 (the glyphs)	`PATTAB`
`>2000-27FF`	plane 1 (enhanced colour 2 and 3)	`PATTAB+>800`
`>2800-2FFF`	plane 2 (enhanced colour 3)	`PATTAB+>1000`
`>3000-35FF`	sprite patterns: plane 0, then +`>200` per plane (64 patterns of 8 x 8 a plane)	`SPTAB`

`>3600-36FF`	the file service's record buffer	
`>3700-37D7`	the file service's PAB	
`>37D8-3FFF`	the disk controller's own: **never touch**	

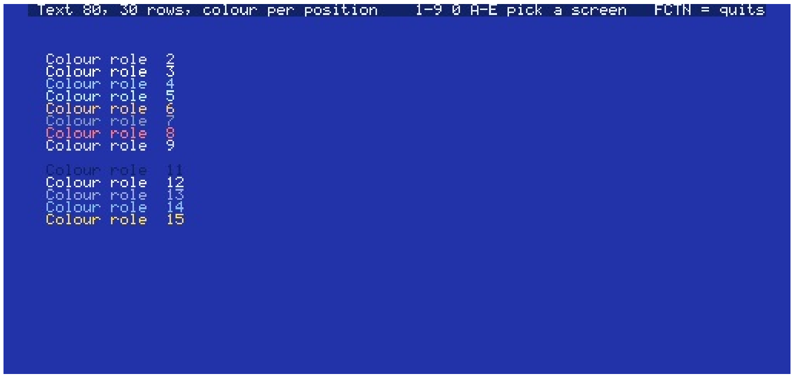
Free VRAM for your own use depends on the mode: with no enhanced colour tiles (`VOTE1` or none) `>2000-2FFF` (4K) is free; with enhanced colour 1 the same; with no sprites `>3000-35FF` (1.5K) is free. The small pieces `>09E0-09FF` and `>1500-17FF` are always free. Tile layer 2's tables and the bitmap layer's bitmap go in these places; the demo puts them at `>2000`.

Graphics II (the standard bitmap map, because the hardware allows the pattern and colour tables only at `>0000` and `>2000`):

Range	Holds	Equate
`>0000-17FF`	patterns (6,144)	`GM2PT`
`>1800-1BBF`	names (768)	`NAMGM2`
`>1C00-1CFF`	sprite attribute table	`GM2SAT`
`>2000-37FF`	colours (6,144)	`GM2CT`
`>3800-3FFF`	sprite patterns	`GM2SPT`

The colour table runs through `>3600-37FF`, which is the file service's buffer and PAB, and the sprite patterns run over the disk controller's area from `>37D8`: **do not use files while Graphics II is showing** unless the bottom of the screen may be damaged. Switch to another mode (or use the bitmap layer, below) to do file I/O.

4. The glyph modes: Text 80, Text 40, Graphics I

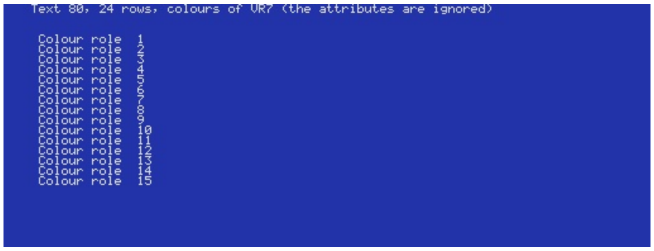


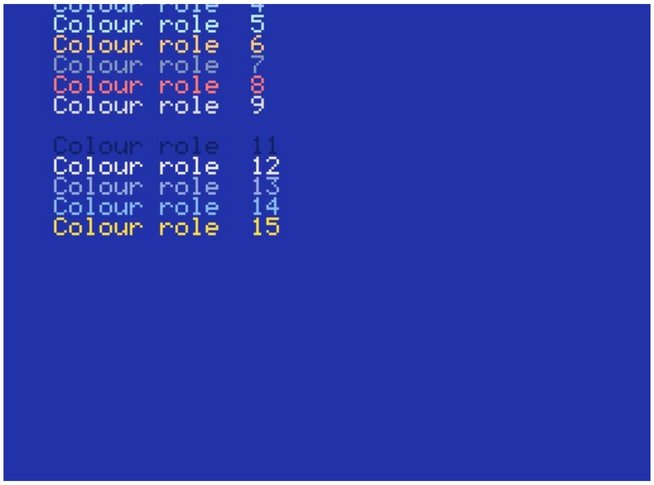
```
 Text 80, 30 rows, colour per position    1-9 0 A-E pick a screen   FCTN = quits


    Colour role  2
    Colour role  3
    Colour role  4
    Colour role  5
    Colour role  6
    Colour role  7
    Colour role  8
    Colour role  9

    Colour role  11
    Colour role  12
    Colour role  13
    Colour role  14
    Colour role  15
```

Text 80 (`VMT80`) is 80 columns of 6 x 8 glyphs. With `VOPOS` every position has an attribute byte (high nibble foreground, low nibble background: palette entries 0-15, which are the kernel's colour *roles*, so a colour scheme recolours the whole screen). Without it all text takes the foreground and background in VR7 (`RTEXT` on `RBG`) and the attributes you pass are ignored. This is the mode Press and the SAMS Kernel's boot use. 24 rows (1,920 cells) or 30 (2,400).

Without `VOPOS`, 24 rows	Text 40, 30 rows, with `VOPOS`
 Text 80, 24 rows, colours of VR7 (the attributes are ignored) Colour role 1 Colour role 2 Colour role 3 Colour role 4 Colour role 5 Colour role 6 Colour role 7 Colour role 8 Colour role 9 Colour role 10 Colour role 11 Colour role 12 Colour role 13 Colour role 14 Colour role 15	Text 40, 30 rows, colour per position Colour role 2 Colour role 3

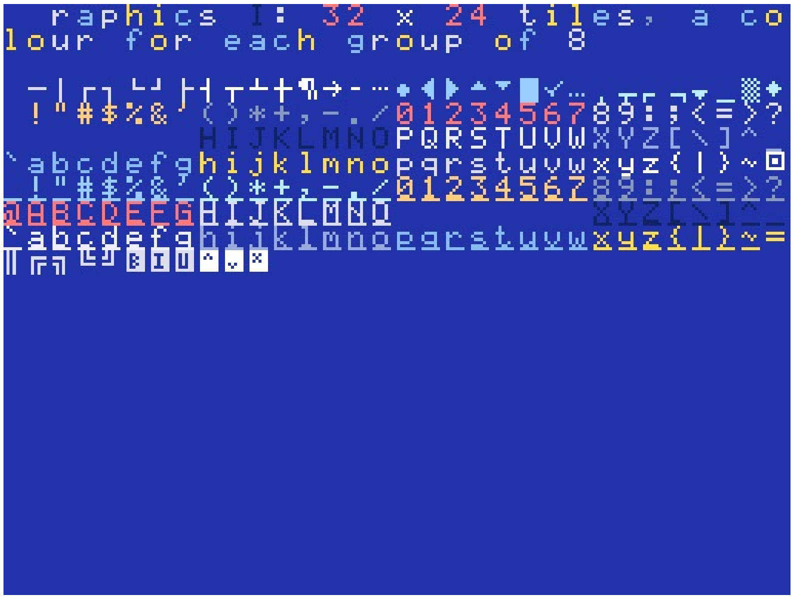


```
Colour role  4
Colour role  5
Colour role  6
Colour role  7
Colour role  8
Colour role  9

Colour role  11
Colour role  12
Colour role  13
Colour role  14
Colour role  15
```

Text 40 is the same with 40 columns, so text and the glyphs look twice the size; the screen has an 8-pixel margin each side that shows the backdrop.

Graphics I (`VMGM1`) is 32 x 24 (or 30) tiles of 8 x 8 pixels, the glyph's full 8 columns. The colour comes from the 32-byte colour table at `GM1CT`: one byte (foreground nibble, background nibble) for each group of 8 consecutive pattern numbers. `VMODE` fills it with `RTEXT` on `RBG`, the same as the other modes; the demo gives each group a colour of its own:



```
  raphics I: 32 x 24 tiles, a co
lour for each group of 8
```

With enhanced colour (`VOTE1-3`) Graphics I uses an attribute table instead (part 5), per tile *code* or, with `VOPOS`, per screen position.

The text routines (`PUTS`, `PUTC`, `FILLN`, `FILLA`, `DECW`, `HEX2`, `HEX4`, `CLS`, `BOX`, `VSHIFT`) work in these three modes and follow `TXCOLS`/`TXROWS`. In a mode with no attribute table (`TXATTR` = 0) they skip the attribute: pass any value.

5. Enhanced colour for tiles (VOTE1, VOTE2, VOTE3)

Normal colour gives a tile pixel one of two colours. In an enhanced colour mode each tile pixel takes **1, 2 or 3 bits** from that many pattern **planes** (the pattern table, plus `>800` and `>1000` further on) and the result, 0 to 1, 3 or 7, selects an entry of a palette *set* chosen by the tile's **attribute byte**:

Attribute bit	Meaning
7 (`>80`)	priority: the tile draws over sprites
6 (`>40`)	flip horizontally
5 (`>20`)	flip vertically

4 (`>10`)	colour 0 transparent (the layer or backdrop behind shows through)
3-0	the palette select

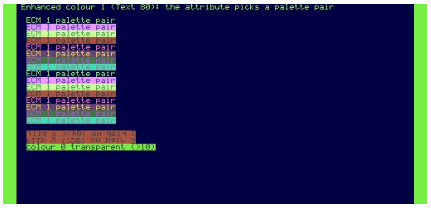
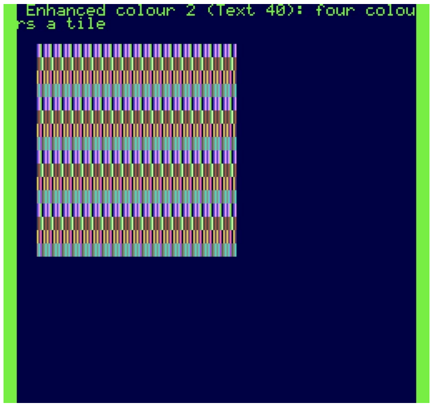
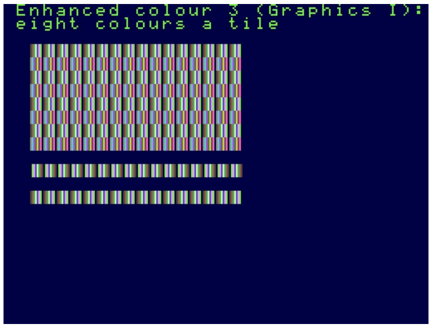
Mode	Planes	Colours	Palette entries used
`VOTE1`	1	2	`(attribute AND >F) * 2`, and +1: sixteen pairs (entries 0-31; VR24's bit 1 adds 32)
`VOTE2`	2	4	`(attribute AND >F) * 4` plus the pixel value: sixteen sets of 4 (0-63)
`VOTE3`	3	8	`(attribute AND >E) * 4` plus the pixel value: eight sets of 8 (0-63; bit 0 ignored)

So in `VOTE3` the attribute `2` picks entries 8-15, `4` picks 16-23 and so on. **The attribute is per tile *code*** (a table of 256 bytes at the colour-table address) unless you add `VOPOS`, when it is **per screen position** (the table at `ATRTAB` that Text 80 uses): `VOPOS` is almost always what you want, and it is what the text routines write (`PUTS`'s attribute argument).

`VMODE` sets the default attribute to 0, loads the scheme into entries 0-15, and for enhanced colour makes **entry 1 the text colour** (it copies the scheme's entry 2 over it), so that a one-plane glyph (pixel 1 = entry 1) shows as text on the page (pixel 0 = entry 0, the scheme's background). Everything else about the palette is yours: load what you need with `F18PAL` (part 9) after `VMODE`. `SETSCH` re-applies the tweak, and rewrites entries 0-15 only.

The glyph set only has plane 0, so ordinary text in an enhanced colour mode is the two colours of pixel 0 and pixel 1. To draw a multi-plane tile, write its planes yourself:

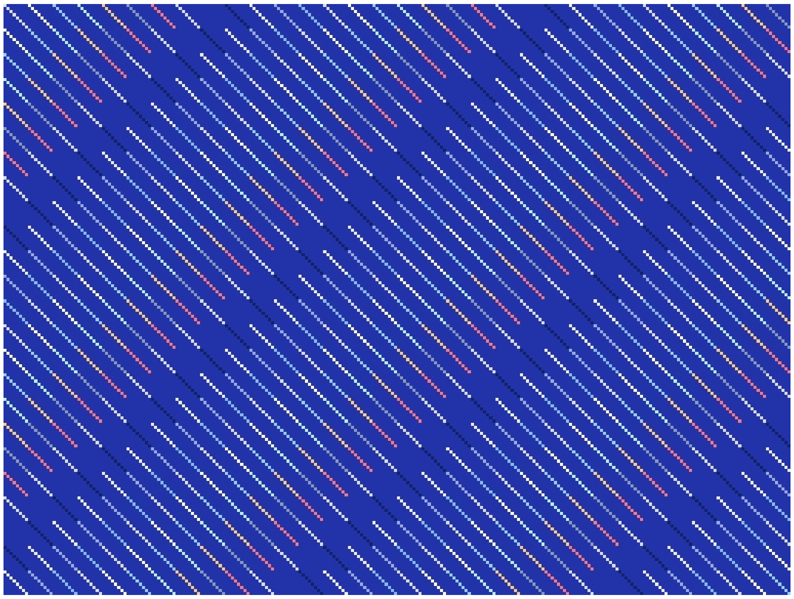
```
        LI   R0,PATTAB+8           glyph 1, plane 0 ...
        LI   R1,>AA00
        LI   R2,8
        BL   @VFILL
        LI   R0,PATTAB+>800+8      ... plane 1 ...
        LI   R1,>CC00
        LI   R2,8
        BL   @VFILL
        LI   R0,PATTAB+>1000+8     ... plane 2
        LI   R1,>F000
        LI   R2,8
        BL   @VFILL                every row: the pixels count 7 6 5 4 3 2 1 0 across the tile
```

Enhanced colour 1, Text 80	Enhanced colour 2, Text 40	Enhanced colour 3, Graphics I
Enhanced colour 1 (Text 80): the attribute picks a palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair ECM 1 palette pair flip x (>40) on pair [illegible] flip y (>20) on pair [illegible] colour 0 transparent (>10)	Enhanced colour 2 (Text 40): four colou rs a tile	Enhanced colour 3 (Graphics I): eight colours a tile

In the left screen each row is a different palette pair (the attribute's low nibble) and the last three rows are the same text with flip x, flip y and colour 0 transparent. In the middle, each row is a tile of four colours in a different set of four; on the right, a tile of eight colours in eight different sets, then flipped in x. Text 80 and Text 40 tiles are 6 pixels wide (the left six columns of each pattern byte); Graphics I tiles are 8.

The attribute you hand to `PUTS`, `FILLA`, `CLS` and `BOX` is this byte in an enhanced colour mode: **do not pass a role attribute such as `(RTEXT*16+RBG)*256`**, whose bits 7-4 mean priority/flip/transparent here (the demo's title bar learned this: it clears the attribute to 0).

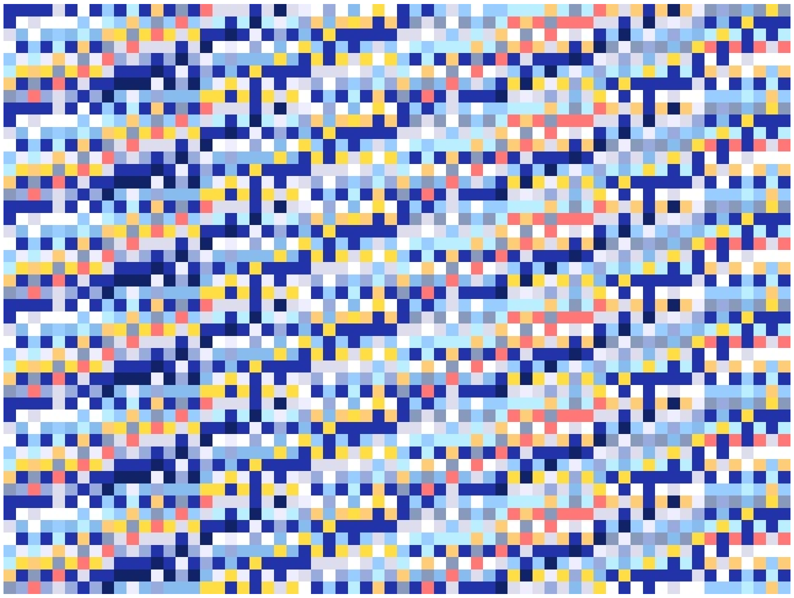
6. Graphics II (bitmap) and Multicolor



Graphics II (`VMGM2`) is a bitmap: 256 x 192 pixels in three vertical thirds of 64 rows. The name table has 768 cells, each a tile number. A tile's 8 pattern bytes (one per pixel row) and 8 colour bytes (foreground nibble, background nibble, one per pixel row, so each row of 8 pixels can have its own two colours) are looked up in the pattern and colour tables **in the third of the screen the cell lies in**. `VMODE` fills the names with `0-255` three times over (the standard arrangement), so the cell at tile row `r`, column `c` (`r` = 0-23) is tile `(r AND 7) * 32 + c`, and the pixel at (x, y) is bit `7 - (x AND 7)` of

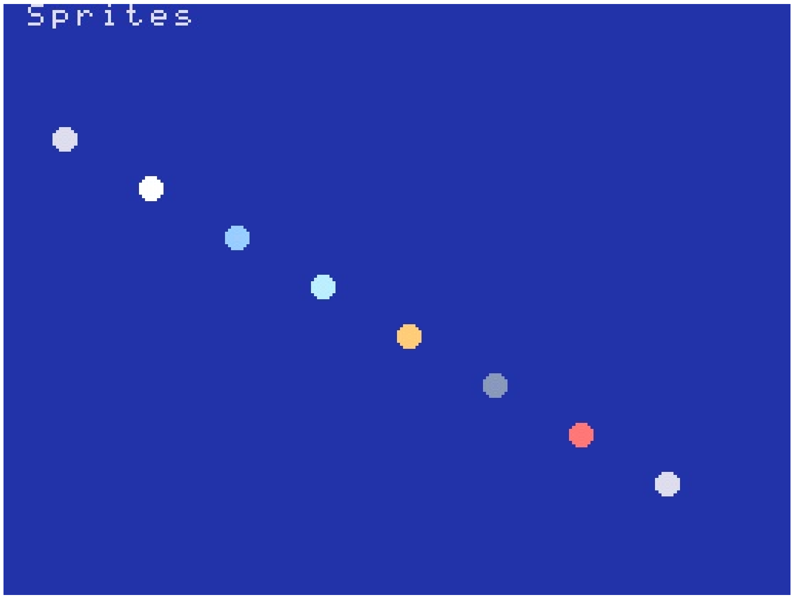
```
        GM2PT + (y>>6)*2048 + (((y>>3) AND 7)*32 + (x>>3))*8 + (y AND 7)
```

with its colours in the byte at the same offset from `GM2CT`. Write patterns and colours with `VMBW`/`VFILL`; the demo gives every tile a diagonal line and a colour of its own. The text routines are not for this mode (the name table holds tile numbers, not text). 24 rows only; **do not use files while it is showing** (part 3).



Multicolor (`VMMC`) is 64 x 48 "pixels" of 4 x 4. Each of the 32 x 24 name-table cells holds a pattern number `n`; the colours of that cell's four blocks are in the **eight bytes at `PATTAB + n*8`**: the cell is 2 blocks wide and 2 high, and the table gives the colours for a row of blocks per byte: byte `k`'s high nibble is the left block and its low nibble the right, where `k = (cell row AND 3) * 2 + (0 for the top row of blocks, 1 for the bottom)`. The pattern number therefore selects four *pairs of rows* at once, and the usual practice (the TMS9918A manual's) is to fill the name table with `(row AND 3) * 32 + column`, so that each of 128 patterns serves one cell column in each group of four rows, and to fill the patterns with the picture; that is what the demo does, with invented colours. Colour 0 is transparent (the backdrop shows). 24 or 30 rows.

7. Sprites



Up to 32 sprites, in every mode (in Text 80/40 only on an unlocked F18A, which the kernel always has). Turn them on with `VOSPR` (VR51 = 32); `VOS16` for 16 x 16, `VOSMG` to double the size. Without `VOSPR` the stop-sprite register (VR51) is 0: nothing is drawn however the table reads. **Hide a sprite with VR51 or by moving it, not with the TMS9918A's `>D0` terminator**: with 30 rows the F18A ignores that terminator.

Sprite attribute table at `SATAB` (`GM2SAT` in Graphics II), 4 bytes per sprite, sprite 0 first (sprite 0 draws in front):

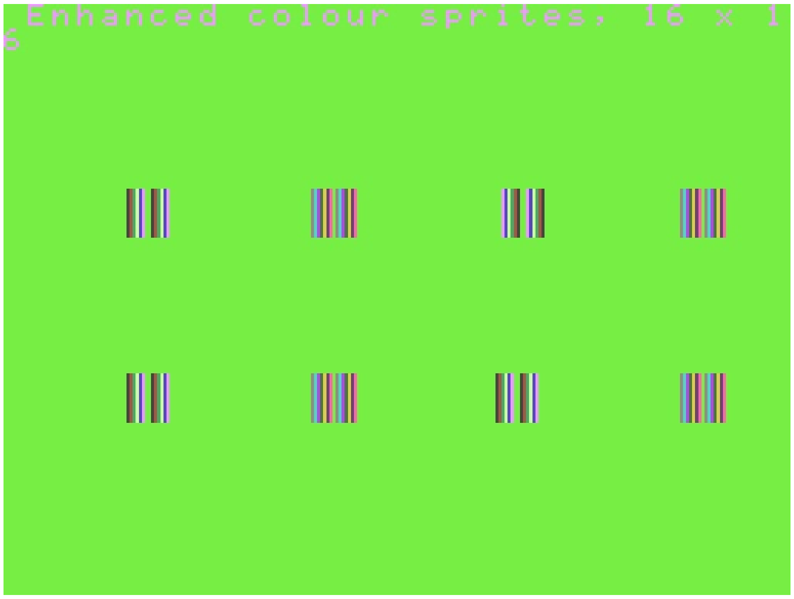
Byte	Meaning
0	Y, one less than the screen row of its top line (the F18A can also use the true Y: VR49 bit 3)
1	X
2	pattern number (a multiple of 4 for a 16 x 16 sprite)
3	bits 3-0 colour (the palette select with enhanced colour); bit 7 early clock (X - 32); bit 6 flip x; bit 5 flip y; bit 4 force 16 x 16 for this sprite

Sprite motion (1.32.0, `"motion": true`: 226 bytes of kernel). The kernel's own mover, because the console's (the interrupt routine adding a velocity to each sprite) is switched off. The program owns a **motion table** in its RAM, 8 bytes a sprite: Y and X as words of 8.8 fixed point (the high byte is the attribute table's own byte), then the Y and X velocities, signed 8.8 dots a frame (`>0100` one dot a frame down or right, `>FF00` one up or left, `>0080` a dot every other frame). `SPMINI` (R0 = the table, R1 = how many sprites, 0-32, R2 = the attribute table's address, `SATAB` or `GM2SAT`) clears the table and starts the motion (a count of 0 stops it); `SPMPUT` (R0 = sprite, R1 = Y, R2 = X in dots) places a sprite and writes its entry; `SPMVEL` (R0 = sprite, R1 = Y velocity, R2 = X velocity) sets how it moves, or the program writes the table directly. **Every frame** (`KFRAME`: the main loop, `KIDLE`, `KWAIT`: so sprites keep moving through a `LINEIN`, a file call or a long computation that polls) each sprite whose velocity is not zero moves and its Y and X bytes in the attribute table are written; a stationary sprite's entry is left alone, so the program may write it itself. Positions wrap at 256 dots as on the console. Not inside `KPFR`, which a program calls in the middle of a stream to VRAM: the step waits (`SPMSKP`) until the next frame. Pattern, colour and the other attribute bytes are the program's; bouncing off an edge, collision and animation are the program's too (`examples/motion/motion.a99`: eight balls turned round at the edges by the module's TICK, 40 lines). The kernel routines use R0-R4: keep a loop's counters elsewhere. **The helpers** (`"motion_helpers": true`, with `"motion"`: 320 bytes more, 7,444 in all) take the common chores off the program. **`SPMBOX`** (R0 = first sprite, R1 = last, R2 = the address of a 4-word box in the program's RAM: Y min, Y max, X min, X max in dots; R2 = 0 turns it off) makes those sprites bounce: after each step a sprite that is past an edge **and still going out** is put back as far inside as it was outside and its velocity on that axis is negated; one that is going in, or placed outside the box and moving inward, is left alone (so it cannot stick). The box is read from RAM every frame, so the program may move its walls; sprites outside the range keep wrapping. Keep the box more than a sprite's speed away from 0 and 255. **`SPMHIT`** (R0, R1 = two sprites, R2 = their size in dots; answer R0 = 1 or 0) says whether they overlap: both their Y and their X are less than that far apart, the screen's wrap taken into account (Y 252 and 3 are 7 apart); a sprite number the table does not hold gives 0. **`SPMSWP`** (R0, R1) exchanges two sprites' velocities, what two equal balls do in a head-on meeting: call it when `SPMHIT` first says they have met (`examples/motion`: eight balls, `SPMBOX` for the walls, `SPMHIT` and `SPMSWP` for each of the 28 pairs, with a flag per pair so that a meeting swaps once). `SPMHIT` tests squares by their corners, not pixels; for the shapes, `lib/spmpix.a99`'s `SPMPIX` compares the patterns pixel for pixel (PROGRAMMERS-GUIDE.md), and `lib/fixed.a99` has sines, directions and square roots for aiming and speeds. The F18A's own coincidence flag is not offered: the kernel reads the video status every frame and clears it. Test group `motion` (17 checks), in the simulator only.

Sprite **patterns** are at `SPTAB` (`GM2SPT` in Graphics II): 8 bytes an 8 x 8 pattern, and a 16 x 16 sprite is four patterns in the order top-left, bottom-left, top-right, bottom-right. The kernel spaces the extra planes of enhanced colour sprites `>200` apart (VR29), so each plane holds 64 patterns of 8 x 8, and the colour of a sprite pixel with `VOSE1-3` is built exactly as for tiles: 1, 2 or 3 bits from the planes, the palette set from the **colour nibble of byte 3**:

Mode	Palette entries used
normal	nibble = the sprite's colour: palette entry `nibble` (plus VR24's sprite select, bits 5-4)
`VOSE1`	`(VR24 AND >20) + nibble*2`, plus the pixel (0-1)
`VOSE2`	`nibble*4`, plus the pixel (0-3)
`VOSE3`	`(nibble AND >E)*4`, plus the pixel (0-7)

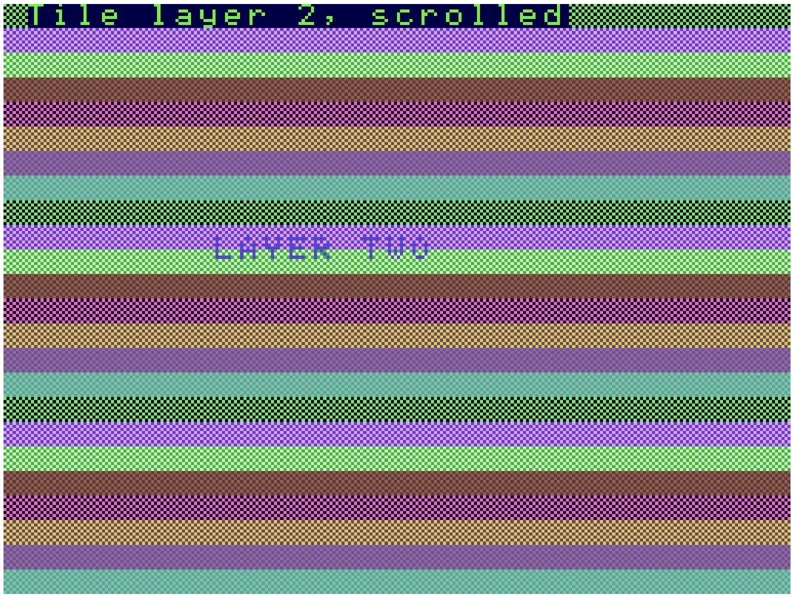
Pixel value 0 is always transparent. The first screenshot above is eight 8 x 8 balls; this is the enhanced colour demo, four 16 x 16 sprites in colour sets 0, 2, 4 (flipped in x) and 6, then sets 12, 10, 8 and 14:



Enhanced colour sprites, 16 x 1
6

Sprite **linking** (VR49 bit 2: a sprite's position and early-clock bit are added to a parent's, set in the second half of the attribute table at `SATAB+>80`), the sprites-per-line limit (VR30: 0 = the jumper, 4 or 32; 31 = all) and the stop sprite (VR51) are set with `VREG`; the demo does not use them. Collision and "fifth sprite" flags are in status register 0 as on the TMS9918A (VR50 bit 3 makes the F18A report the maximum instead).

8. Tile layer 2, hardware scrolling, page sizes



The F18A has **two tile layers**. Layer 1 is the screen `VMODE` sets up; layer 2 is a second screen with its own name table (`VR_NAME2`, 1K units) and colour/attribute table (`VR_COLR2`, 64-byte units), the same pattern table, its own scroll registers and, with `VR49` bit 7, drawn *over* layer 1 (and over the sprites unless VR50 bit 0 is set). Where its pixels are transparent (an enhanced colour tile with the attribute's bit 4 set, colour 0) layer 1 shows. The demo is Graphics I with enhanced colour 1: layer 1 is a coloured checker, layer 2 the words LAYER TWO on transparent tiles:

```
        LI   R0,VR_NAME2           layer 2's name table at >2000 (1K units)
        LI   R1,>0800
        BL   @VREG
        LI   R0,VR_COLR2           its attribute table at >2400 (64-byte units: >90)
        LI   R1,>9000
        BL   @VREG
        LI   R0,VR_ECM             tile layer 2 on, with tile ECM 1 as VMODE had set it
        LI   R1,>9000
        BL   @VREG
```

Scrolling. Each layer scrolls by pixel, independently: VR27 / VR28 (layer 1 horizontal / vertical), VR25 / VR26 (layer 2). In Text 80 the horizontal value counts the 256-pixel scale, so each unit moves 2 pixels of the 512-pixel-wide screen. The scrolling wraps at the edge of the screen (of the margin-less area, in the text modes). The **page size** bits of VR29 (bits 1 and 0 for layer 1, bits 5 and 4 for layer 2) make the name table two pages wide and/or two high (the wrap then goes to the next page: `>400`/`>800` further for a horizontal/vertical page) so that a map larger than the screen can scroll; set the layer's name table at the first of the pages. VR50 bit 4 turns layer 1 off entirely.

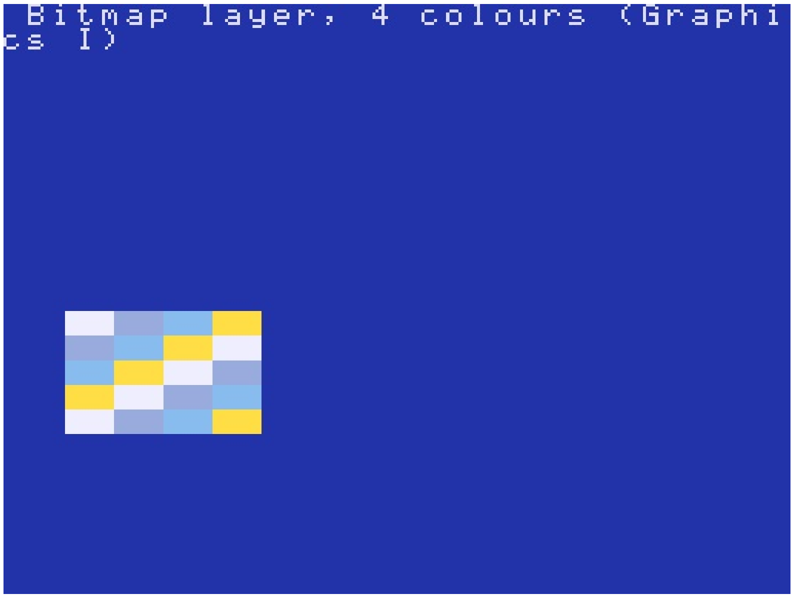
9. Palette

The F18A has 64 palette entries of 12 bits (`>0RGB`). Entries 0-15 hold the kernel's roles; the power-up palette also fills 16-63 with variations. `F18PAL` loads entries:

```
        LI   R0,16                the first entry
        LI   R1,MYPAL             a table of words >0RGB
        LI   R2,48                how many
        BL   @F18PAL
```

VR24 selects which 16 (or 32, 64) entries each layer uses in the modes without their own selection: bits 1-0 for tile layer 1 (times 16), bits 3-2 for tile layer 2, bits 5-4 for sprites (normal mode). In the normal colour modes colour 0 draws nothing (the backdrop, entry VR7's low nibble, shows through); in enhanced colour that is the attribute's bit 4.

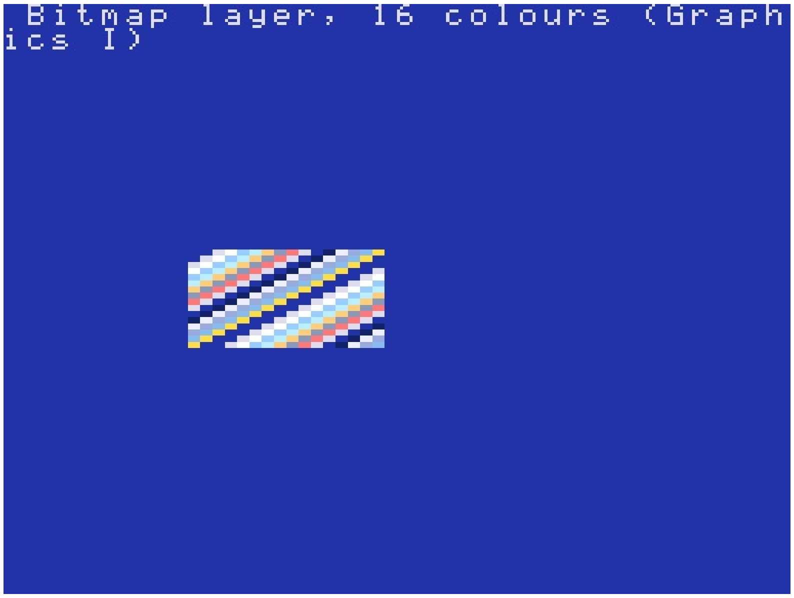
10. The bitmap layer


Bitmap layer, 4 colours (Graphi
cs I)

A rectangular bitmap, drawn over the tile layers (below the sprites and tile layer 2): its pixels take their colour from palette entries chosen by VR31, from 4 colours (2 bits a pixel) or, with the "fat pixel" bit, 16 colours (4 bits a pixel, each pixel two columns wide).

Register	**Meaning**
`VR_BMPCT` (31)	bit 7 on; bit 6 priority (draws over non-background tiles); bit 5 colour 0 transparent; bit 4 fat pixels; bits 3-0 palette select (x4 in 4-colour mode; bits 3-2 only, x16, in fat mode)
`VR_BMPAD` (32)	the bitmap's address in VRAM, in 64-byte units
`VR_BMPX`, `VR_BMPY` (33, 34)	the rectangle's top left
`VR_BMPW`, `VR_BMPH` (35, 36)	its width (0 = 256) and height in pixels

The bitmap holds the pixels row by row, **4 pixels to a byte in both forms** (four 2-bit pixels, or two 4-bit pixels each two columns wide), so a rectangle of w x h pixels takes w x h / 4 bytes: 64 x 40 is 640 bytes. Without the priority bit the layer shows only where the tile layer is the background colour. The demo's first screen above is a 64 x 40 four-colour bitmap with palette entries 12-15; the second is 16 colours:



```
 Bitmap layer, 16 colours (Graph
ics I)
```

This is the way to put real graphics on a screen without giving up the file service (Graphics II uses the controller's VRAM). **The demo uses it over Graphics I.**

Pictures: `tools/skpic.py` and `lib/picture.a99` (1.30.0, from TXA's picture format and tools, MIT). `skpic.py` makes a **picture record** from a PNG or a TI Artist picture (`NAME_P` and `NAME_C`, 6,144 bytes each, cropped to 96 rows from row 48 or squeezed to half height as TXA does, by `NAME_T` beside it): a PNG of at most 4 colours becomes **format 2** (256 x rows, 2 bits a pixel, 4 colours) and any other **format 4** (128 x rows fat pixels, 4 bits a pixel, 16 colours); the record is a header, the palette (`>0RGB` words) and the pixels run-length packed. An `app.json` data entry whose file ends `.skp` is built that way: `{"file": "harbour.skp", "from": "pics/harbour.png"}` (or `"from": "pics/HARBOUR"` for TI Artist's pair; `"rows": 96` for the whole band), and the build names its first SAMS page `PD_HARBOUR`. A module that `COPY`s `picture.a99` calls **`PICSHOW`** (R0 = the page, R1 = the VRAM address for the pixels, a multiple of 64, R2 = the top scanline, R3 = the bytes of VRAM free there; R0 = 0 shown, 1 not a picture, 2 does not fit, 3 bad address): it decodes the pixels into VRAM, loads the palette into F18A entries 32-, sets VR31-VR36 and turns the layer on; **`BMLOFF`** and **`BMLON`** switch it without touching the pixels. **Rows: a row is 64 bytes in both formats, so `>2000-35FF` (the free run in Text 80 with no enhanced colour tiles and no sprite patterns) holds 88 rows exactly**, which is why `skpic.py` defaults to 88 (TXA's 96 rows, 6,144 bytes, reach `>37FF` into the file service's PAB and buffer; a program that uses no files, or redraws the picture after each, can give it that). The picture is below `>3600`, so file calls do not disturb it. Test group `pic`: the decoded pixels equal the reference decoder's byte for byte, the palette, the registers, the windows put back, `BMLOFF`/`BMLON`, the three failure answers, the TI Artist route and the build. **Not seen on a real F18A, and over Text 80 only on js99er (see below).**

Over Text 80 (reported by the TXA project, 1 November 2026, tested on js99er only; I have not run it, and no real F18A has been tried). The layer's position test is in 256-pixel units, but js99er's offset into the bitmap is not halved to match, so js99er draws such a picture **doubled horizontally** (a bug reported to js99er's author); what the hardware does is unknown. TXA's working settings for a 16-colour (fat pixel) picture above the 80-column text, with the bitmap at `>2800`:

Register	Value	Meaning
`VR_BMPAD` (32)	`>A0`	the bitmap at `>2800` (64-byte units)
`VR_BMPX` (33), `VR_BMPY` (34)	`>00`, `>08`	x 0; y 8: under an 8-scanline status row
`VR_BMPW` (35), `VR_BMPH` (36)	`>00`, `>60`	256 wide: the full Text 80 width; 96 rows
`VR_BMPCT` (31)	`>D8`	on, priority, fat pixels (16 colours), palette select 8: entries 32-47

The picture's 16 colours go into palette entries 32-47, so the text's colours are untouched; the text rows the picture covers are blanked and the text window starts below it.

Keep a bitmap out of every file buffer. The disk controller reads and writes through VRAM, and so does this kernel's file service: a bitmap stored where a file call puts its data is overwritten by the next disk call (the same warning as for Graphics II's colour table, part 6). In the map above `>3600-36FF` is the file service's record buffer, `>3700-37D7` its PABs and `>37D8-3FFF` the controller's. **The free run in Text 80 is `>2000-35FF`, 5,632 bytes** (and only `>2000-2FFF` with sprites or enhanced colour planes), so TXA's 6,144-byte bitmap at `>2800` overlaps the file service and has to be redrawn after every disk call (TXA's `PICRDW`). Either keep a picture to 5,632 bytes (for example 128 x 88 fat pixels) or redraw it after file calls.

11. Other registers (set them with VREG)

Register	Meaning
`VR_PALCT` (47)	palette data port mode (used by `F18PAL`); leave it
`VR_INCR` (48)	the VRAM address increment after each data port access (signed; 1 normally)
`VR_FLAGS` (50)	bit 7 reset all registers; bits 6, 5 GPU triggers on horizontal / vertical sync; bit 4 layer 1 off; bit 3 report the sprite maximum; bit 2 scan lines; bit 1 position attributes; bit 0 layer 2 vs sprite priority
`VR_HINT` (19)	the scan line of a horizontal interrupt (0 = off)
`VR_SPRMX` (30)	sprites per line
`VR_STOP` (51)	the stop sprite
`VR_GPUHI/LO`, `VR_GPUCT` (54-56)	the F18A's GPU: start address and control (the kernel uses it for `VSHIFT`; a program of your own should keep out of `GPUAD`)
`VR_STAT` (15)	the status register that reads next

The F18A is also a 9918A-compatible: VR0-VR7 mean what they do on the 9918A, and an *unlocked* F18A (the kernel keeps it unlocked) accepts registers 8-63.

12. What `VMODE` does not cover, and what I did not check

- **Graphics II with 30 rows** is not offered (the pattern thirds do not cover rows 25-30); Multicolor with 30 rows is set but not drawn by the demo.
- **Text 80's bitmap layer** is untested (part 10). **Linking, horizontal interrupt, the scan lines, the GPU beyond `VSHIFT`, page sizes** are available through `VREG` but the kernel's tests do not exercise them.
- **The renderer is not the F18A.** It reproduces js99er's drawing, which I read and ported, and the demo's colours and positions are checked against it. A real F18A may differ in details; confirm each mode you rely on on the hardware (`examples/modes` is the program to run: keys `1-9 0 A-E`).
- **PAL and 50 Hz, the PICO9918:** the F18A registers above are those of the F18A. The kernel treats a PICO9918 as an F18A, but I did not test its modes.
- **Sound, joysticks** are not video.

13. Trying the modes

```
python3 tools/skbuild.py examples/modes                 build/modes/videomodes8.bin: load it in js99er (F18A, 512K SAMS)
python3 tools/f18render.py build/modes/videomodes8.bin shot.png "<WAIT 5>6<WAIT 20>" 900     key 6: Graphics II, drawn to
shot.png
python3 tools/mkshots.py                                the fifteen images in docs/images
python3 tests/test_kernel.py modes modesquit            the pixel checks, and the register restore on quit
```

Keys in the demo: `1-4` the four text screens, `5` Graphics I, `6` Graphics II, `7` Multicolor, `8` `9` `0` the three enhanced colour levels, `A` `B` sprites, `C` tile layer 2, `D` `E` the bitmap layer; `FCTN =` quits. In your own tests, `f18render.render(m)` returns the picture as a numpy array; `f18render.palette_rgb(m)` the 64 colours; compare a pixel with `pal[entry]`, as `tests/test_kernel.py` does.